CONTROL SYSTEMS PROJECT (ARICRAFT PITCH SYSTEM)

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Abstract

In this project, we present the analysis and modeling of the aircraft pitch control system using both Simulink and MATLAB. One of the goals of this project is to develop the dynamics of the model. In this model we try to captures the pitch behavior of an aircraft and design a suitable controller to regulate the pitch angle in response to changes in input. This system is built on the basis of Newton's second law, then linearized and applied to the assumptions of small scatterings. A transfer function is derived and verified using a Simulink model. Then we analyzed the stability by using the Routh table, which showed no change in the signals, proving that this system is stable. A PID controller is designed using the Root Locus method to meet specified performance criteria such as minimal overshoot, fast settling time, and zero steady-state error. Simulation results confirm the effectiveness of the designed controller in achieving the desired dynamic response while maintaining robustness and stability under various conditions.

Introduction

Aircraft pitch control is a critical aspect of flight dynamics **(fig.1)**, as it governs the aircraft's rotation about its lateral axis, directly affecting its ascent, descent, and level flight. In real-world applications, any disturbance—whether from turbulence, payload shift, or control input—can alter the pitch angle, potentially compromising flight stability and safety. Therefore, a reliable and responsive pitch control system is essential to maintain desired flight trajectories and ensure passenger comfort and safety.

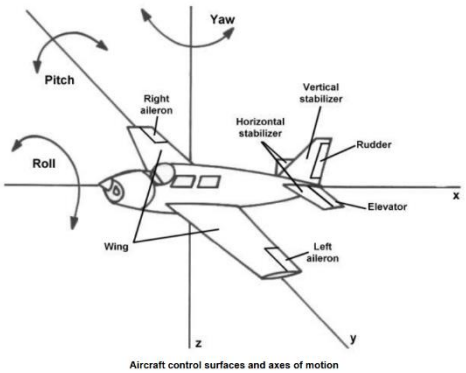


Figure 1: Aircraft Control Structures

In this project, we aim to design a control system that regulates the aircraft's pitch using a PID controller. Two primary methods are employed: Root Locus analysis and Simulink PID Tuner.

The Root Locus technique offers a graphical method to analyze how the closed-loop poles of a system change with variations in the controller gain. It enables the manual design of a PID controller that satisfies key dynamic response characteristics, such as reduced overshoot, improved settling time, and minimal steady-state error. This method helps in understanding how proportional, integral, and derivative actions influence system behavior and where the poles must be placed to achieve desired performance.

The Simulink PID Tuner, on the other hand, provides an interactive and automated environment to fine-tune PID parameters. It allows real-time visualization of the closed-loop step response, giving designers the flexibility to optimize performance criteria like rise time, overshoot, and robustness. This tuning method is especially useful for quick deployment and precise adjustments without manually analyzing root locus plots.

Through both analytical and simulation-based approaches, this project demonstrates the full control system development workflow—from physical modeling to controller design and system evaluation—on an aircraft pitch control application.

System Modelling and Properties

1) Governing Differential Equations:

Firstly, we will use the newton's second law for both linear and angular momentum as referenced to an inertial frame:

$$\Sigma F = \frac{d}{dt}(mv)$$

$$\Sigma M = \frac{d}{dt}H$$

Then we will use this equation as a vector equation:

$$\Sigma F_x = \frac{d}{dt}(mu)$$

$$\Sigma F_y = \frac{d}{dt}(mv)$$

$$\Sigma F_z = \frac{d}{dt}(mw)$$

We have the force components of (aerodynamic, propulsive and gravitational)
Then we can express the momentum equations to:

$$\mathrm{L} = \frac{d}{dt}(H_x)$$

$$\mathrm{M} = \frac{d}{dt}(H_y)$$

$$\mathrm{N} = \frac{d}{dt}(H_z)$$

Where:

- L is the component of momentum in X
- M is the component of momentum in Y
- N is the component of momentum in Z
- H_x is the moment of momentum in X
- H_y is the moment of momentum in Y
- H_z is the moment of momentum in Z

Then we take a small element of mass, velocity and force ($\delta m, v \; and \; \delta F$):

$$\sum F = \delta m \frac{dv}{dt}$$

The total external force which is the sum of the all-elemental forces on the mass:

$$\Sigma \delta F = F$$

The velocity can be expressed as the summation of the velocity centre of the airplane mass and the elemental velocity relative to the centre mass velocity

$$v = v_c + \frac{dr}{dt}$$

Then use the elemental in the in newton's second law:

$$\sum \delta F = F = \frac{d}{dt}\sum \left(v_c + \frac{dr}{dt}\right)\delta m$$

If the mass is constant:

$$F = m\frac{dv_c}{dt} + \frac{d}{dt}\sum\frac{dr}{dt}\delta m$$

$$F = m\frac{dv_c}{dt} + \frac{d^2}{dd^2}\sum r\delta m$$

Then the value of $\sum r\delta m = 0$ as it is measured from the centre of mass.

The angular velocity can be expressed as:

$$\omega = pi + qj + rk$$

$$r = xi + yj + zk$$

Then:

$$F = m\frac{dv_c}{dt}$$

Then we can use a relation relating the moment to the moving airplane and the moment equation becomes:

$$\sum M = \frac{d}{dt}\delta H = \frac{d}{dt}(rv)\delta m$$

Then the summation of velocities can be:

$$v = v_c + \frac{dr}{dt} = v_c + \omega r$$

Where:

- ω is the angular velocity
- r is the position from centre of mass

Then the total moment can be:

$$H = \sum \delta H = \sum (rv_c)\delta m + \sum (r(\omega r))\delta m$$

If v_c is constant then and use the angular velocity in a vector so H can be:

$$H = (pi + qj + rk)\sum (x^2 + y^2 + z^2)\delta m - \sum (xi + yj + zk)(px + qy + rz)\delta m$$

Then the moment of momentum express as:

$$H_x = p\sum (y^2 + z^2)\delta m - q\sum xy\delta m - \sum xy\delta m$$

$$H_y = -p\sum xy\delta m - q\sum (x^2 + z^2)\delta m - r\sum yz\delta m$$

$$H_z = -p\sum xz\delta m - q\sum yz\delta m - r\sum (x^2 + y^2)\delta m$$

Then find the summation that represent the airplane mass moment of inertia:

$$I_x = \iiint (y^2 + z^2)\delta m \qquad I_y = \iiint (x^2 + z^2)\delta m \qquad I_z = \iiint (x^2 + y^2)\delta m$$

$$I_{xy} = \iiint xy\ \delta m \qquad I_{xz} = \iiint xz\ \delta m \qquad I_{yz} = \iiint yz\ \delta m$$

We can express the moment of momentum as:

$$H_x = pI_x - qI_{xy} - rI_{xz}$$

$$H_y = -pI_{xy} + qI_y - rI_{yz}$$

$$H_z = pI_{xz} - qI_{yz} + rI_z$$

For simplicity we should fix the axis system and then we should not deal with the varying moments and products of moments with time for rotating body:

$$\left.\frac{dA}{dt}\right|_I = \left.\frac{dA}{dt}\right|_B + \omega A$$

Where:

- I and B is the inertial and body fixed frame of reference

Then after applied this to the force and the momentum:

$$F = m\left.\frac{dv_c}{dt}\right|_B + m(\omega H)$$

$$M = \left.\frac{dH}{dt}\right|_B + \omega H$$

Then we can assume the symmetrical XZ plane ($I_{yz} = I_{xy} = 0$) then the momentum can be:

$$L = I_x P_{dot} - I_{xy} r_{dot} + qr\left(I_z - I_y\right) - I_{xz} Pq$$

$$M = I_x P_{dot} - rp(I_x - I_z) + I_{xy}(P^2 - r^2)$$

$$N = -I_{xz} P_{dot} + I_z r_{dot} + Pq\left(I_y - I_x\right) + I_{xz} qr$$

Gravitational and thrust forces are a huge influence on the dynamics of the airplane. As commonly, gravitational force acts on the airplane centre of gravity. Since the body axis system is fixed to the centre of gravity, gravitational force does not produce any moments. Although, it will contribute to external forces acting on the plane.

$$(F_x)_{gravity} = -mg \sin\theta$$

$$(F_x)_{gravity} = -mg \cos\theta \sin\phi$$

$$(F_x)_{gravity} = -mg \cos\theta \cos\phi$$

Then the Propulsive forces and moments acting along the body axis:

$$(F_x)_{propulsive} = X_T \qquad (F_y)_{propulsive} = Y_T \qquad (F_z)_{propulsive} = Z_T$$

$$(L)_{propulsive} = L_T \qquad (M)_{propulsive} = M_T \qquad (N)_{propulsive} = N_T$$

Applying small disturbance theory, the linearized equations relate aerodynamic forces and moments to key variables, with the elevator as the main control. See the table for a summary:

Longitudinal Equations

$$\left(\frac{\boldsymbol{d}}{\boldsymbol{dt}} - \boldsymbol{x_u}\right)\Delta\boldsymbol{u} - \boldsymbol{X_w}\Delta\boldsymbol{W} + (\boldsymbol{g\ cos\ \theta_0})\Delta\boldsymbol{\theta} = \boldsymbol{X_{\delta e}}\Delta\boldsymbol{\delta_e} + \boldsymbol{X_{\delta r}}\,\Delta\boldsymbol{\delta_r}$$

$$-Z_u\Delta_u + \left[(1 - Z_\omega)\frac{d}{dt} - Z_\omega\right]\Delta\omega - \left[(u_0 + Z_q)dt - g\sin\theta_0\right]\Delta\theta = Z_{\delta e}\Delta\delta_e + Z_{\delta r}\Delta\delta_r$$

$$-M_u\Delta u - \left(M_\omega\frac{d}{dt} - M_\omega\right)\Delta\omega + \left(\frac{d^2}{dt^2} - M_q\frac{d}{dt}\right)\Delta\theta = M_{\delta e}\Delta\delta_e + M_{\delta r}\Delta\delta_r$$

Lateral equations

$$\left(\frac{\boldsymbol{d}}{\boldsymbol{dt}} - \boldsymbol{Y_v}\right)\Delta\boldsymbol{v} - \boldsymbol{Y_p}\Delta\boldsymbol{p} + (\boldsymbol{u_0} - \boldsymbol{Y_r})\Delta\boldsymbol{r} - (\boldsymbol{g\ cos\ \theta_0})\Delta\boldsymbol{\phi} = \boldsymbol{Y_{\delta e}}\Delta\boldsymbol{\delta_e}$$

$$-L_v\Delta_v + \left(\frac{d}{dt} - L_p\right)\Delta p - \left(\frac{I_z}{I_x}\frac{d}{dt} + L_r\right)\Delta r = L_{\delta e}\Delta\delta_e + L_{\delta r}\Delta\delta_r$$

$$-N_v\Delta v - \left(\frac{I_x}{I_z}\frac{d}{dt} + N_p\right)\Delta p + \left(\frac{d}{dt} - N_r\right)\Delta r = N_{\delta e}\Delta\delta_e + N_{\delta r}\Delta\delta_r$$

Using small disturbances' theory and by adding perturbation equations, we come up with these equations describing the aircraft motion:

- Rolling rate: $p = \dot{\phi} - \dot{\varphi}\sin\theta$
- Yawing rate: $q = \dot{\theta}_{dot}\cos\phi + \dot{\varphi}_{dot}\cos\theta\sin\phi$
- Pitching rate: $r = \dot{\varphi}\cos\theta\sin\phi - \dot{\theta}_{dot}\sin\phi$
- Pitch angle: $\dot{\theta}_{dot} = q\cos\phi - r\sin\phi$
- Roll angle: $\sin\phi = p + q\sin\phi\tan\theta + r\cos\phi\tan\theta$
- Yaw angle: $\dot{\varphi}_{dot} = (q\sin\phi + r\cos\phi)\sec\theta$

Assume steady cruise with constant velocity and altitude Aircraft motion in the vertical direction:

$$L - W = m\frac{dV}{dt}$$

$$W = mg$$

Where:

- L is Lift
- W is Weight
- m is Aircraft mass
- g is Gravitational acceleration
- V is Aircraft Vertical Velocity
- $\frac{dV}{dt}$ is Rate of change of the vertical velocity with time

Then the lift:

$$L = \frac{1}{2}V^2 S C_L$$

$$C_L = C_L(\alpha)$$

Where:

- S is wing area
- C_L is Coefficient of Lift

Rotation about the lateral axis of the aircraft is influenced by pitch moment and the aerodynamic forces on the horizontal tail:

$$M - M_a = I_{yy}\frac{d\omega}{dt}$$

$$M = \frac{1}{2}\rho V^2 S c C_m$$

$$C_M = C_M(\alpha, \theta)$$

$$M_a = \frac{1}{2}\rho V^2 S_t c C_{M,t}(x_t - x_{ac})$$

Where:

- M is Pitch Moment
- M_a is Aerodynamic Forces on the horizontal tail
- ω is Angular velocity in the pitch direction
- $\frac{d\omega}{dt}$ is Rate of change of the angular velocity
- I_{yy} is Moment of inertia about the lateral axis
- C IS Mean Aerodynamic wing chord
- $C_M$is Pitching moment coefficient
- S_t is Horizontal Tail Area
- $C_{M,t}$ is Pitching moment Coefficient of the tail
- x_t is Distance from the centre of gravity to the aerodynamic centre of the tail
- x_{ac} is Distance from the wing leading edge to the wing aerodynamic centre

Assume the angle of attack = the pitch angle so:

$$C_L = C_L(\theta)$$

$$C_M = C_M(\theta)$$

We will assume that the aircraft is in steady-cruise at constant altitude and velocity; thus, the thrust, drag, weight and lift forces balance each other in the *x*- and *y*-directions. We will also assume that a change in pitch angle will not change the speed of the aircraft under any circumstance (unrealistic but simplifies the problem a bit). Under these assumptions, the longitudinal equations of motion for the aircraft can be written as follows.

$$\dot{\alpha} = \mu\Omega\sigma\left[-(C_L + C_D)\alpha + \frac{1}{(\mu - C_L)}q - (C_W \sin\gamma)\theta + C_L\right]$$

$$\dot{q} = \frac{\mu\Omega}{2i_{yy}}\left[[C_M - \eta(C_L + C_D)]\alpha + [C_M + \sigma C_M(1 - \mu C_L)]q + (\eta C_W \sin\gamma)\delta\right]$$

$$\dot{\theta} = \Omega q$$

From Boeing's commercial aircraft:

$$\dot{\alpha} = -0.313\alpha + 56.7q + 0.232\delta$$

$$\dot{q} = -0.0139\alpha - 0.426q + 0.0203\delta$$

$$\dot{\theta} = 56.7q$$

To find the transfer function of the above system, we need to take the Laplace transform of the above modelling equations:

$$sA(s) = -0.313A(s) + 56.7Q(s) + 0.232\Delta(s)$$

$$sQ(s) = -0.0139A(s) - 0.426Q(s) + 0.0203\Delta(s)$$

$$s\theta(s) = 56.7Q(s)$$

After few steps of algebra, you should obtain the following transfer function:

$$P(s) = \frac{\theta(s)}{\Delta(s)} = \frac{1.151s + 0.1774}{s^3 + 0.739s^2 + 0.921s}$$

2) System's Parameters:

- $\boldsymbol{\alpha}$ **Definition**: Angle of attack **Unit**: rad or deg
- $\boldsymbol{q}$ **Definition**: Pitch rate **Unit**: $\frac{rad}{s}$
- $\boldsymbol{\theta}$ **Definition**: Pitch angle **Unit**: rad or deg
- $\boldsymbol{\delta}$ **Definition**: Elevator deflection angle **Unit**: rad or deg
- $\boldsymbol{\mu}$ $= \frac{\boldsymbol{\rho S\bar{c}}}{\boldsymbol{4m}}$ **Unit**: dimensionless
- $\boldsymbol{\rho}$ **Definition**: Density of air **Unit**: $\frac{kg}{m^3}$
- $\boldsymbol{S}$ **Definition**: Wing platform Area **Unit**: m^2
- $\boldsymbol{\bar{c}}$ **Definition**: Average chord length **Unit**:
- $\boldsymbol{m}$ **Definition**: Aircraft Mass **Unit**: kg
- Ω $= \frac{\boldsymbol{2U}}{\boldsymbol{\bar{c}}}$ **Unit**: s^{-1}
- $\boldsymbol{U}$ **Definition**: Equilibrium flight speed **Unit**: $\frac{m}{s}$
- $\boldsymbol{C_T}$ **Definition**: Coefficient of thrust **Unit**: dimensionless
- $\boldsymbol{C_D}$ **Definition**: Coefficient of drag **Unit**: dimensionless
- $\boldsymbol{C_L}$ **Definition**: Coefficient of lift **Unit**: dimensionless
- $\boldsymbol{C_W}$ **Definition**: Coefficient of weight **Unit**: dimensionless
- $\boldsymbol{C_M}$ **Definition**: Coefficient of pitch moment **Unit**: dimensionless
- $\boldsymbol{\gamma}$ **Definition**: Flight path angle **Unit**: rad or deg
- $\boldsymbol{\sigma}$ $= \frac{\boldsymbol{1}}{\boldsymbol{1+\mu C_L}} = \boldsymbol{const.}$ **Unit**: dimensionless
- $\boldsymbol{i_{yy}}$ **Definition**: Normalized moment of inertia **Unit**: dimensionless

- η $= \mu\sigma C_M = const.$ **Unit**: dimensionless

3) Simulink Model:

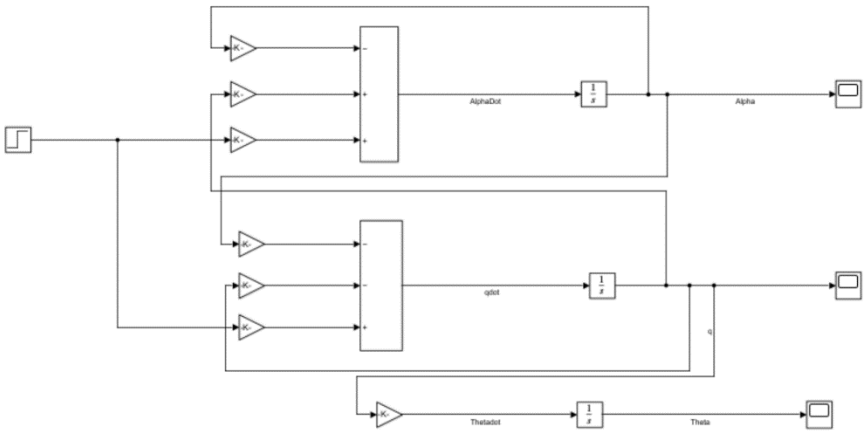


Figure 2: Simulink Model

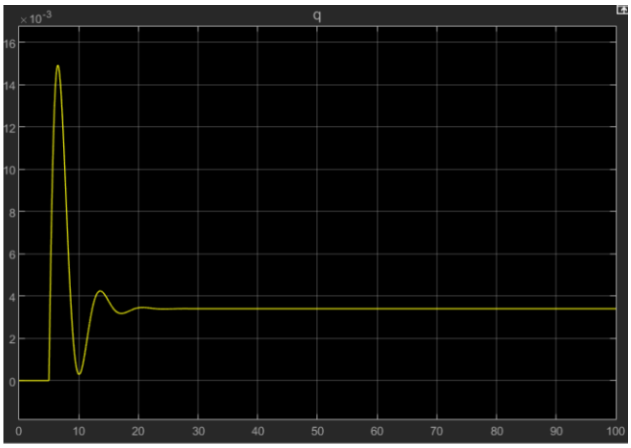


Figure 3: Plotted q Using Simulink Model

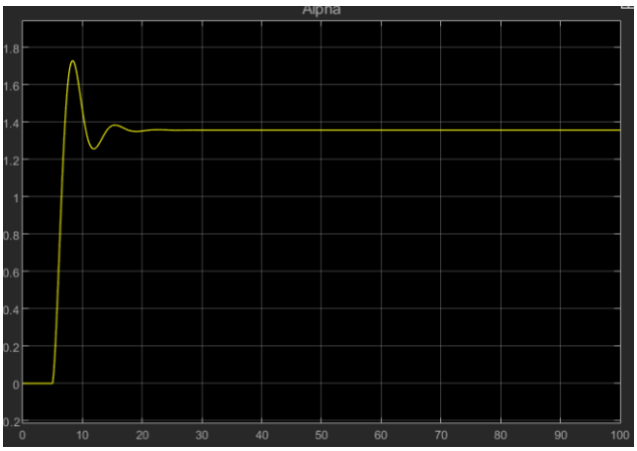


Figure 4: Plotted alpha Using Simulink Model

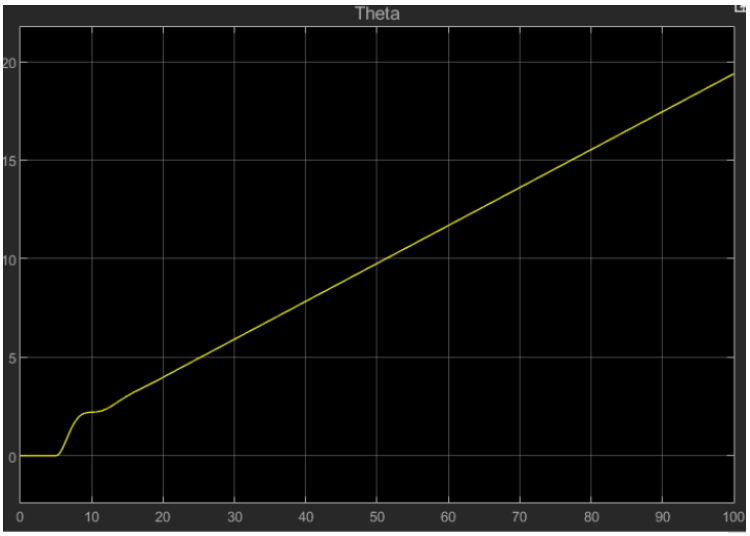


Figure 5: Plotted theta Using Simulink Model

4) Open-Loop Transfer Function:

$$P(s) = \frac{\theta(s)}{\Delta(s)} = \frac{1.151s + 0.1774}{s^3 + 0.739s^2 + 0.921s}$$

5) System's Stability:

Power of *s*	**First Column**	**Second Column**
s^3	1	0.921
s^2	0.739	0
s^1	0.921	0
s^0	0	

The system is stable because the all signs are positive the roots ([1,0.739,0.921,0])

$$0.0000 + 0.0000i$$

$$-0.3695 + 0.8857i$$

$$-0.3695 - 0.8857i$$

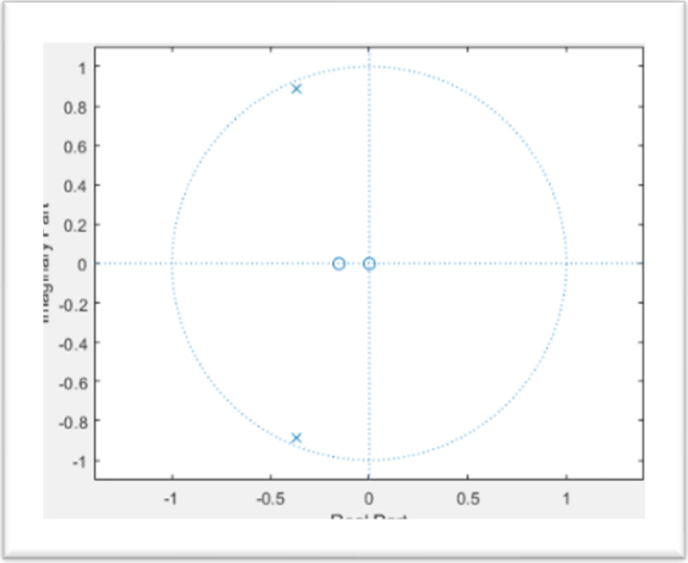


Figure 6: System-Representing Root Locus

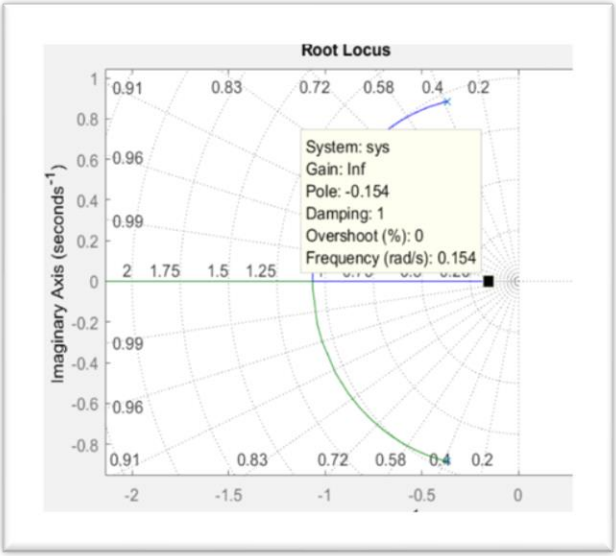
Controller Design

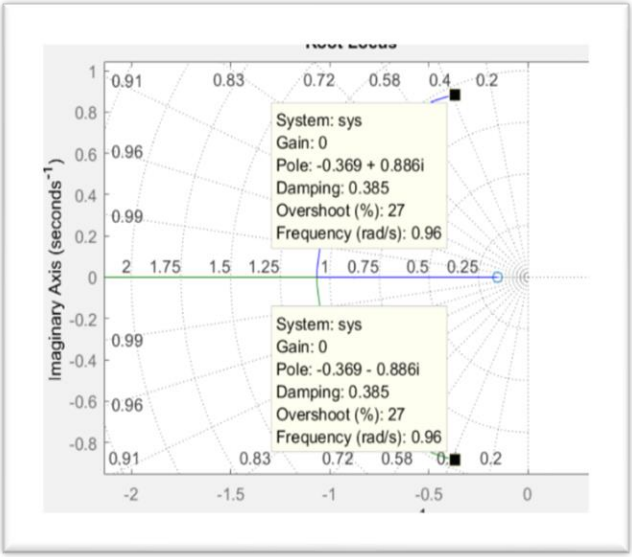
1) Closed-Loop Response Objectives:

1) The target to make the overshoot less than ten percent to avoid the oscillation when fast response
2) Hope to make the settling time less than 5 sec to make pitch angle stable fast after the disturbance
3) Hope to find the rise time that make the system respond in a fast way to the input
4) Hope to make the steady state error zero for the step input to make accurate tracking of pitch
5) Hope to maintain stability of the system in all condition avoiding oscillation
6) Hope to minimize the influence of any external disturbance

2) PID Controller Using Root Locus:

Firstly, we lunched the muddling of the system and define the characteristics like stability, transient response and the steady state error after that we using the root locus analysis and select the proportional, integral and derivative gain to demonstrate the behaviour of the closed loop system after that we define the transfer function:





3) PID Controller Using Simulink:

The PID controller was used to control the closed-loop response of the aircraft pitch system. Moreover, the PID Tuner was utilized to fine-tune the controller parameters, ensuring optimal performance in terms of overshoot, settling time, and steady-state error.

The modelling in the next image contains:

1- Step input
2- Disturbance
3- PID Tuner Block
4- State-Space Block. The variables were set as:

$$A = \begin{bmatrix} -0.313 & 56.7 & 0 \\ -0.0139 & -0.426 & 0 \\ 0 & 56.7 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0.232 \\ 0.0203 \\ 0 \end{bmatrix} \qquad C = eye(3) \qquad D = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

5- Scope

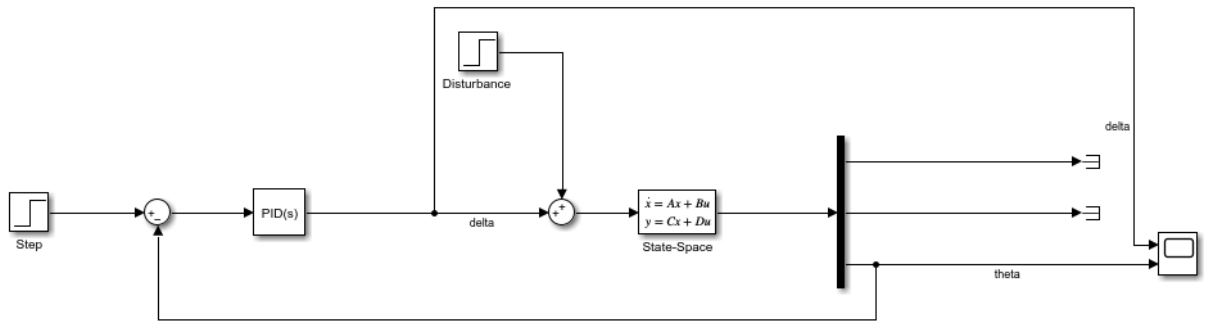


Figure 7: PID Controller Diagram Using Simulink

Simulation Results and Analysis

1) Step Input, Open-Loop Response:

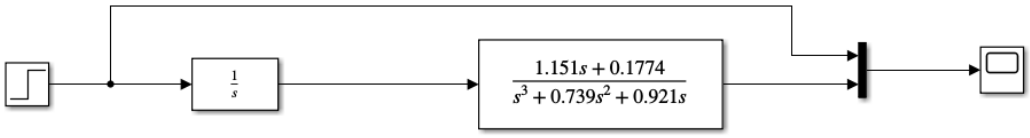


Figure 8: Open-Loop Representing Diagrams

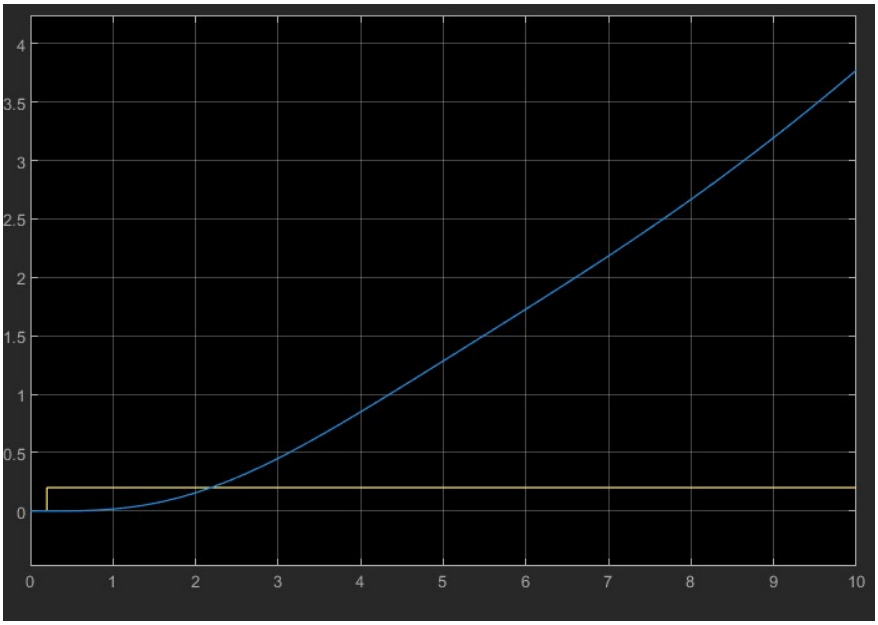


Figure 9: Open-Loop System Response

- The input signal is plotted by the yellow line, while the system response is plotted by the blue line.
- The open-loop response to a unit step input shows an unstable behavior characterized by a positive-feedback behavior, where the system increases to infinity.
- This emphasizes the need for a controller to regulate the pitch dynamics.

2) Closed-Loop Response For Different Step Inputs:

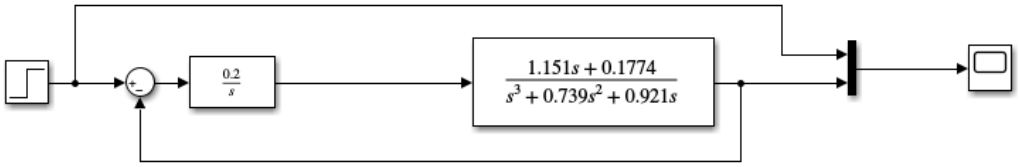


Figure 10: Closed-Loop Representing Diagrams

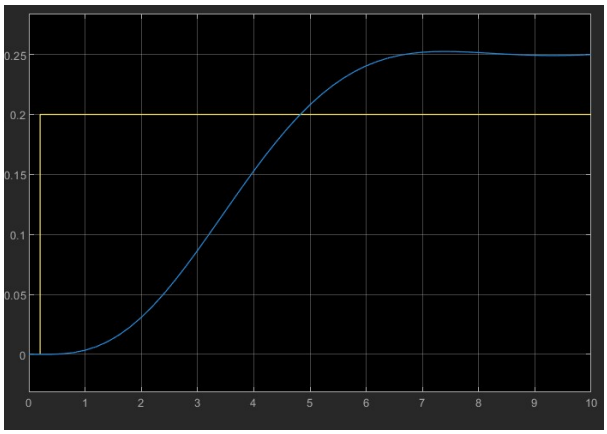


Figure 11: Response for step input 0.2/s

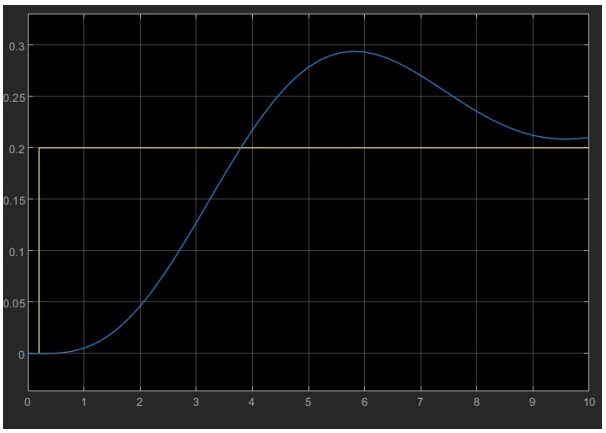


Figure 12: Response for step input 0.3/s

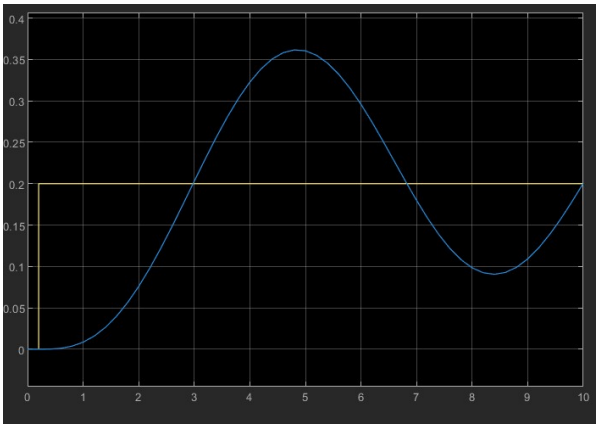


Figure 13: Response for step input 0.5/s

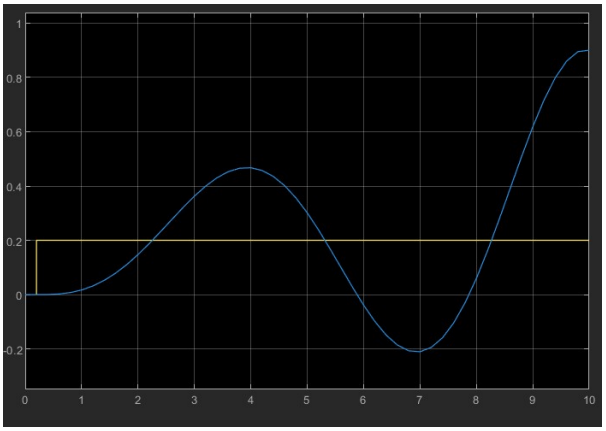


Figure 14: Response for step input 1/s

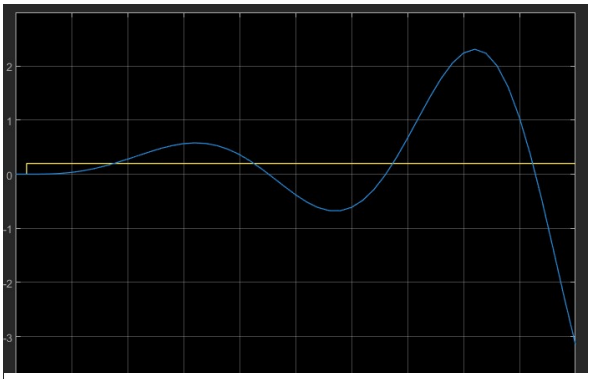


Figure 15: Response for step input 2/s

- $\frac{0.2}{s}$ Step input: The system has a slow but stable rise. Settling occurs around 5 seconds. This is a good indication that the controller handles low-amplitude reference with precision.
- $\frac{0.3}{s}$ Step input: the system has a faster response than $\frac{0.2}{s}$ step input. The system also has a noticeable overshoot.
- It can be observed by increasing the step input, the amplitude and overshoot increase.
- Step input $\frac{1}{s}$ has moderate characteristics compared to the other step inputs.

3) Tuned Closed-Loop Response:

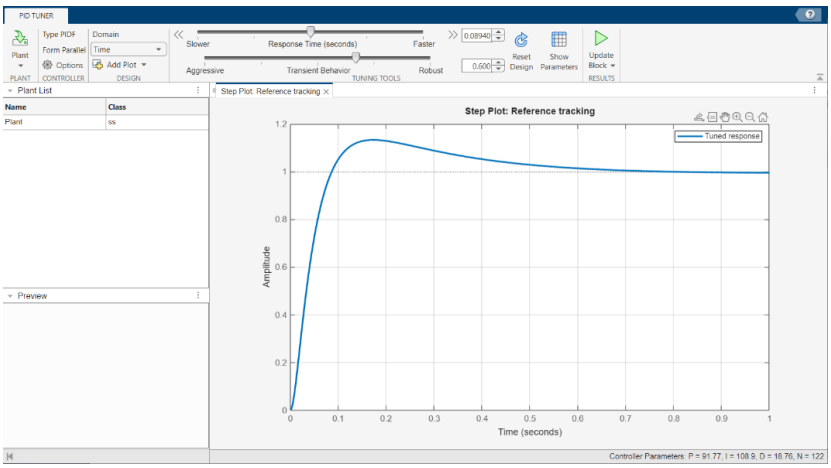


Figure 16: The System Response Using PID Automatic Tuner

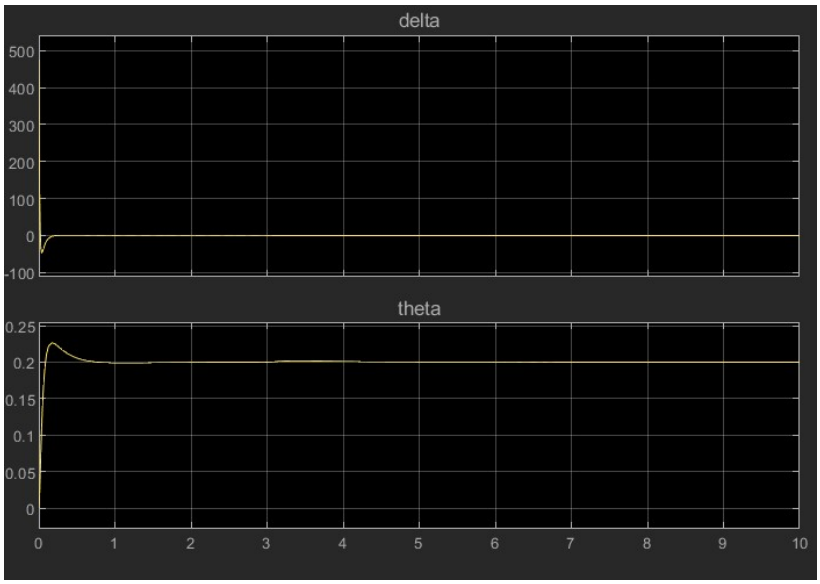


Figure 17: The Delta and Theta Responses using Simulink PID Tuner

- The tuned PID controller results in a smoother and more optimized pitch response.
    - The overshoot is significantly reduced (below 5%)
    - The system settling time is within 2–3 seconds across various step inputs.
    - The rise time has improved without introducing oscillations
    - The steady-state error has been eliminated.
- This confirms that the PID Tuner effectively optimized controller gains to meet the desired performance criteria.

Conclusion

This project modelled and controlled an aircraft pitch system using control theory and Simulink. Starting from Newton's laws, the system's differential equations were derived and linearized using small disturbance theory. Key parameters were defined, and a Simulink model was built. The open-loop transfer function was obtained, and Routh-Hurwitz analysis confirmed the system's stability. A PID controller was designed using Root Locus to meet performance goals: low overshoot, fast settling, and zero steady-state error. Simulations showed that the tuned controller improved the system's response and maintained stability. The project successfully demonstrated the full process of modelling, controller design, and validation for an aircraft pitch system.

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